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Review

# Learning from class-imbalanced data: Review of methods and applications

Guo Haixiang[a,b,c,1,*], Li Yijing[a,b,1,*], Jennifer Shang[d,*], Gu Mingyun[a], Huang Yuanyue[a], Gong Bing[e]

[a] College of Economics and Management, China University of Geosciences, Wuhan 430074, China
[b] Research Center for Digital Business Management, China University of Geosciences, Wuhan 430074, China
[c] Mineral Resource Strategy and Policy Research Center of China University of Geosciences(WUHAN), Wuhan 43007, China
[d] The Joseph M. Katz Graduate School of Business, University of Pittsburgh, Pittsburgh, PA 15260, USA
[e] Department of Industrial Engineering, Business Administration and Statistic, E.T.S Industrial Engineering, Universidad Politécnica de Madrid, C/José Gutiérrez Abascal, 2- 20086, Madrid, Spain

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ABSTRACT

Rare events, especially those that could potentially negatively impact society, often require humans' decision-making responses. Detecting rare events can be viewed as a prediction task in data mining and machine learning communities. As these events are rarely observed in daily life, the prediction task suffers from a lack of balanced data. In this paper, we provide an in depth review of rare event detection from an imbalanced learning perspective. Five hundred and seventeen related papers that have been published in the past decade were collected for the study. The initial statistics suggested that rare events detection and imbalanced learning are concerned across a wide range of research areas from management science to engineering. We reviewed all collected papers from both a technical and a practical point of view. Modeling methods discussed include techniques such as data preprocessing, classification algorithms and model evaluation. For applications, we first provide a comprehensive taxonomy of the existing application domains of imbalanced learning, and then we detail the applications for each category. Finally, some suggestions from the reviewed papers are incorporated with our experiences and judgments to offer further research directions for the imbalanced learning and rare event detection fields.

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## 1. Introduction

Rare events, unusual patterns and abnormal behavior are difficult to detect, but often require responses from various management functions in a timely manner. By definition, rare events refer to events that occur much less frequently than commonly occurring events (Maalouf and Trafalis, 2011). Examples of rare events include software defects (Rodriguez et al., 2014), natural disasters (Maalouf and Trafalis, 2011), cancer gene expressions (Yu et al., 2012), fraudulent credit card transactions (Panigrahi et al., 2009), and telecommunications fraud (Olszewski, 2012).

In the field of data mining, detecting events is a prediction problem, or, typically, a data classification problem. Rare events are difficult to detect because of their infrequency and casualness; however, misclassifying rare events can result in heavy costs. For financial fraud detection, invalid transactions may only emerge out of hundreds of thousands of transaction records, but failing to identify a serious fraudulent transaction would cause enormous losses. The scarce occurrences of rare events impair the detection task to imbalanced data classification problem. Imbalanced data refers to a dataset within which one or some of the classes have a much greater number of examples than the others. The most prevalent class is called the majority class, while the rarest class is called the minority class (Li et al., 2016c). Although data mining approaches have been widely used to build classification models to guide commercial and managerial decision-making, classifying imbalanced data significantly challenges these traditional classification models. As were discussed on existing surveys, the reasons are fivefold:

* Corresponding authors.
*E-mail addresses:* faterdumk0732@sina.com (G. Haixiang), liyijing024@hotmail.com, liyijing024@gmail.com (L. Yijing), shang@katz.pitt.edu (J. Shang), 550312686@qq.com (G. Mingyun), huangyuanyue1991@126.com (H. Yuanyue), gongbing1112@gmail.com (G. Bing).

[1] Guo Haixiang and Li Yijing contributed equally in this paper.

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(1) Standard classifiers such as logistic regression, Support Vector Machine (SVM) and decision tree are suitable for balanced training sets. When facing imbalanced scenarios, these models often provide suboptimal classification results, i.e. a good coverage of the majority examples, whereas the minority examples are distorted (López et al., 2013).
(2) The learning process guided by global performance metrics such as prediction accuracy induces a bias towards the majority class, while the rare episodes remain unknown even if the prediction model produces a high overall precision (Loyola-González et al., 2016). Some original discussion can be found in Weiss and Hirsh (2000) and Weiss (2004).
(3) Rare minority examples may possibly be treated as noise by the learning model. Contrarily, noise may be wrongly identified as minority examples, since both of them are rare patterns in the data space (Beyan and Fisher, 2015).
(4) Even though skewed sample distributions are not always difficult to learn (such as when the classes are separable), minority examples usually overlap with other regions where the prior probabilities of both classes are almost equal. Denil and Trappenberg (2010) has discussed overlapping problem under imbalanced scenario.
(5) Besides, small disjuncts (Jo and Japkowicz, 2004), a lack of density and small sample size with high feature dimensionality (Wasikowski and Chen, 2010) are challenges to imbalanced learning, which often cause learning models to fail in detecting rare patterns (Branco et al., 2016; López et al., 2013).

Many machine learning approaches have been developed in the past decade to cope with imbalanced data classification, most of which have been based on sample techniques, cost sensitive learning and ensemble methods (Galar et al., 2012; Krawczyk et al., 2014; Loyola-González et al., 2016). There is also one book in this area, see He and Ma (2013). Although several surveys related to imbalanced learning have been published (Branco et al., 2016; Fernández et al., 2013; Galar et al., 2012; He and Garcia, 2009; López et al., 2012; Sun et al., 2009), all of them focused on detailed techniques while application literature is neglected. For researchers from management, biology or other domains, rather than sophisticated algorithms, the problems that can be solved using imbalanced learning techniques and the building of imbalanced learning systems with mature yet effective methods may be of more concern.

In this paper, we aim to provide a thorough overview of the classification of imbalanced data that includes both techniques and applications. At the technical level, we introduce common approaches to deal with imbalanced learning and propose a general framework within which each algorithm can be placed. This framework is a unified data mining model and includes preprocessing, classification and evaluation. At the practical level, we review 162 papers that tried to build specific systems to tackle rare pattern detection problems and develop a taxonomy of existing imbalanced learning application domains. The existing application literature covers most research fields from medical to industry to management.

The rest of this paper is organized as follows. Section 2 describes the research methodology for this study, along with the initial statistics regarding recent trends in imbalanced learning. Section 3 presents approaches to address both binary and multiple class imbalanced data. In Section 4, we first categorize the existing imbalanced learning application literature into 13 domains and then introduce the respective research frameworks. In Section 5, we discuss our thoughts on future imbalanced learning research directions from both a technical and practical point of view. Finally, Section 6 present the conclusions of this paper.

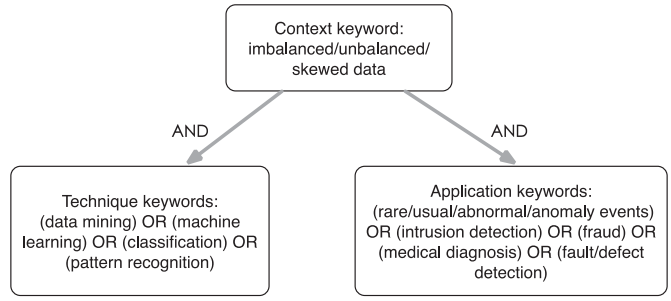


**Fig. 1.** Two-stage keywords tree structure (we use "/" to separate synonyms of the same concept).

## 2. Research methodology and initial statistics

### 2.1. Research methodology

For this study, based on the research methodology of Govindan and Jepsen (2016), a two-stage search procedure was conducted to compile relevant papers published from 2006 to October 2016. In the initial phase, seven library databases which covered most natural science and social science research fields were used to search for and collect literature: Elsevier, IEEExplore, Springer, ACM, Cambridge, Wiley and Sage. Full text search was used and the search term was designed following the search process outlined in Fahimnia et al. (2015). A two-level keywords tree was constructed to provide a comprehensive set of search terms to capture technical and application articles on rare events and imbalanced learning. Fig. 1 presents the search terms for this study. The search phase on the first level was restricted to imbalanced /unbalanced /skewed data, as the focus was on imbalanced data classification. The second level search terms were divided in twofold to cover both technical and practical articles. For techniques, keywords that referred to data mining approaches were used and for practical applications, key words were used that focused on event detection and prediction that included rare events, usual events, abnormal events, defective procedures, fraud, disease, and intrusion. Note that the corresponding inflected forms of a word and the synonyms (such as "anomaly" for "abnormal", "fraudulence" for "fraud") were also considered. The initial search yielded 657 papers, which were downloaded for the next filtering process.

After reviewing each paper manually, 464 papers were found to be relevant to this study. The second stage search was performed during the review and the relevant cross-references were searched for using Google Scholar. At this stage, attempts were made to access all cross-references in Google Scholar or were included in the accessible library databases. After the second stage, 63 papers were added to our review. Therefore, a total of 527 papers were included in this study.

### 2.2. Initial statistics

In this section, we present the initial statistics for the trends in imbalanced learning. Fig. 2 shows the publishing trends by plotting the quantity of publications from 2006 to 2016. A relatively stable growth in the number of publications can be observed after 2006. The only downward trend in the 2011/2013 interval was made up immediately by a sharp rise in the number of publications in 2013–2016. This trend suggests imbalanced learning hitherto remains a valuable research topic.

The initial statistics also show that 192 journals and conference proceedings published a total of 527 papers. The contribution of each journal was counted and the top 20 journals/conferences are shown in Fig. 3. These journals covered 43.5% of all published

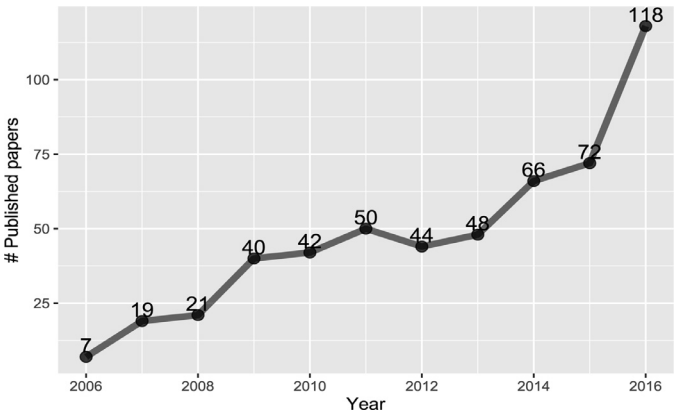


**Fig. 2.** Imbalanced learning publishing trends.

papers in the past decade, most of which are influential journals in the fields of computer science, operations research, management science, engineering and biotechnology. This also illustrates that imbalanced learning research has included both learning techniques from the computer science community and practical applications across a broad range of domains from natural science to management science.

We also collected the titles of all the papers involved in this study and generated a word cloud to capture the most studied imbalanced learning topics. To build the word cloud presented in Fig. 4, we first removed the stop words most commonly used in English such as "the" and "and", then NLTK tool was employed to lemmatize each word. As our goal was to uncover detailed topics in these imbalanced learning papers, some general and frequently appearing words were also eliminated.[1] Fig. 4 shows some of the specific techniques for the imbalanced data classification: re-sampling methods ("over-sampling", "sampling", "under-sampling", etc.), machine learning approaches ("cost-sensitve", "support vector machine", "neural network", "ensemble", etc.) and expert systems ("rule", "system", "fuzzy", etc.). However, some concrete application domains are also shown in the word cloud. Typical word clusters covered several domains, such as "patient", "fraud detection", "telecommunications", and "credit card", which implied that the main categories for the applied papers may lie in fields such as financial management, medical diagnosis, and telecommunications.

## 3. Imbalanced data classification approaches

Hundreds of algorithms have been proposed in the past decade to address imbalanced data classification problems. In this section, we give an overview of the state-of-the-art imbalanced learning techniques. These techniques are discussed in line with the basic machine learning model framework. In Section 3.1, two basic strategies for addressing imbalanced learning are introduced, which are, preprocessing and cost-sensitive learning. Preprocessing approaches include resampling methods conducted in the sample space and feature selection methods that optimize the feature space. The strategies introduced in Section 3.1 are then integrated into the classification models described in Section 3.2. The classifiers are further divided into ensemble classifiers and algorithmic modified classifiers. Section 3.3 discusses multi-class classification as a special imbalanced learning issue to clarify the extension of these binary classification algorithms into a multi-class case. In Section 3.4, the metrics for evaluating and selecting the models are introduced.

### 3.1. Basic strategies for dealing with imbalanced learning

#### 3.1.1. Preprocessing techniques

Preprocessing is often performed before building learning model so as to attain better input data. Considering the representation spaces of data, two classical techniques are often employed as preprocessor:

*3.1.1.1. Resampling.* Resampling techniques are used to rebalance the sample space for an imbalanced dataset in order to alleviate the effect of the skewed class distribution in the learning process. Resampling methods are more versatile because they are independent of the selected classifier (López et al., 2013). Resampling techniques fall into three groups depending on the method used to balance the class distribution:

- Over-sampling methods: eliminating the harms of skewed distribution by creating new minority class samples. Two widely-used methods to create the synthetic minority samples are randomly duplicating the minority samples and SMOTE (Chawla et al., 2002).
- Under-sampling methods: eliminating the harms of skewed distribution by discarding the intrinsic samples in the majority class. The simplest yet most effective method is Random Under-Sampling (RUS), which involved the random elimination of majority class examples (Tahir et al., 2009).
- Hybrid methods: these are a combination of the over-sampling method and the under-sampling method.

We found 156 reviewed papers have utilized resampling techniques, accounting for 29.6% of all papers reviewed. This indicates that resampling is a popular strategy for dealing with imbalanced data. Under-sampling methods are employed 39 times, whereas over-sampling methods are used 84 times; and hybrid-sampling methods are chosen 33 times. Instead using existing re-sampling methods, 52 papers developed novel re-sampling methods. As cluster-based methods (e.g. k-means), distance based methods (e.g. nearest neighbors) and evolutionary methods (e.g. generic algorithm) were the most frequently used strategies to generate or eliminate examples, these have been summarized in Table 1. For method that did not employ the above strategies, we give a brief introduction in the last column.

It should be noted that all of the re-sampling methods allow to re-sample to any desired ratio, and it is not necessary and sufficient to exactly balance the number of majority and minority classes. Zhou (2013) recommended different sample ratio for different data size, Napierala and Stefanowski (2015) studied types of minority class examples and their influences on learning classifiers from imbalanced data. Some papers tried to automatically decide best sampling rate for different Imbalanced Ratios (IR) and problem settings (Lu et al., 2016; Moreo et al., 2016; Ren et al., 2016a; Tomek, 1976; Yun et al., 2016; Yeh et al., 2016; Zhang et al., 2016a).

Three papers studied the performances of different re-sampling methods (Loyola-González et al., 2016; Napierala and Stefanowski, 2015; Zhou 2013). There are some insights from these papers: 1). When there are hundreds of minority observations in the dataset, an under-sampling method was superior to an over-sampling method in terms of computational time. 2). When there are only a few dozen minority instances, the over-sampling method SMOTE was found to be a better choice. 3). If the training sample size is too large, a combination of SMOTE and under-sampling is suggested as an alternative. 4). SMOTE is slightly more effective in recognizing outliers.

[1] These words are "class-imbalanced", "imbalanced", "classification", "class-unbalanced", "unbalanced", "approaches", "approach", "based", "using", "data-sets", "datasets", "sets", "data".

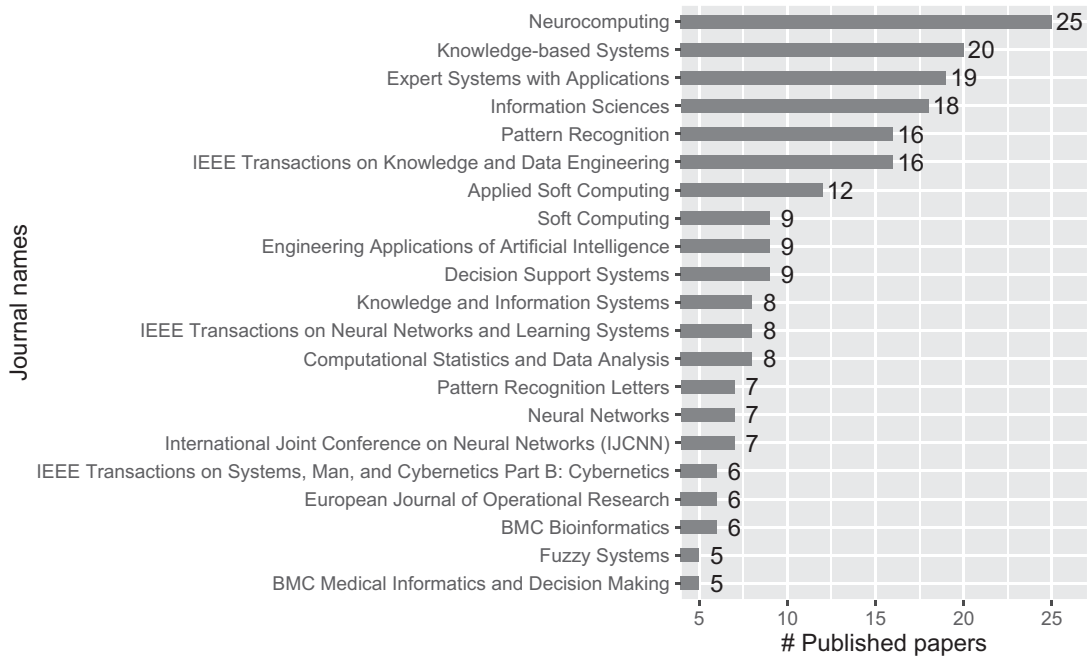


**Fig. 3.** Top 20 journals/conferences in which most imbalanced learning papers were published.

**Fig. 4.** Word cloud of the title words from the collected papers.

*3.1.1.2. Feature selection and extraction.* Compared to the progress achieved in resampling methods, significantly fewer papers considered feature selection. Under imbalanced scenarios, minority class samples can easily be discarded as noise; however, if the irrelevant features in the feature space are removed, this risk is reduced (Li et al., 2016c).

The goal of feature selection, in general, is to select a subset of k features from the entire feature space that allows a classifier to achieve optimal performance, where k is a user-specified or adaptively selected parameter. Feature selection can be divided into filters, wrappers, and embedded methods (Guyon and Elisseeff, 2003). The advantages and disadvantages of these methods can be found in Saeys et al. (2007).

Another way to deal with dimensionality is feature extraction. Feature extraction is related to dimensionality reduction, which transforms data into a low-dimensional space. It should be noted, however, that feature selection techniques are different from feature extraction. Feature extraction creates new features from the original features using functional mapping, whereas feature selection returns a subset of the original features (Motoda and Liu, 2002). There are a variety of techniques for feature extraction, such as Principal Component Analysis (PCA), Singular Value Decomposition (SVD), and f Non-negative Matrix Factorization (NMF) (see Hartmann, 2004). Feature extraction methods tend to be employed more frequently for unstructured data such as images, text and speech.

Table 2 shows 31 reviewed articles employed feature selection or feature extraction. It was found that filter and wrapper feature selection methods were most regularly used. For filter methods, various metrics were used to rank the features, and for wrapper

**Table 1**
Summary of articles employing re-sampling methods.

| Category | Strategy | Articles | Basic ideas behind the novel algorithms |
|---|---|---|---|
| **Over-sampling** | Cluster-based | Nekooeimehr and Lai-Yuen (2016), Cao et al. (2014) | |
| | Distance based (e.g. SMOTE and a modified version) | 15 papers such as Li et al. (2016a), Yun et al. (2016), Zhang et al. (2015a), García et al. (2012), Zhai et al. (2015) | |
| | Evolutionary based | Li et al. (2016a), Yang et al. (2009) | |
| | Other novel method | Chetchotsak et al. (2015) | Created new data with a self-organizing map |
| | | Menardi and Torelli (2014) | Bootstrap form of random over-sampling using a kernel method |
| | | Das et al. (2015) | Used joint probability distribution of data attributes and Gibbs sampling to generate new minority class samples |
| | | Cao et al. (2013) | Over-sampling time series data. |
| **Under-sampling** | Cluster-based | 6 papers such as Sun et al. (2015), Li et al. (2013a), Kumar et al. (2014) | |
| | Distance based | D'Addabbo and Maglietta (2015), Anand et al. (2010) | |
| | Evolutionary based | 6 papers such as Ha and Lee (2016), Galar et al. (2013), Krawczyk et al. (2014) | |
| **Hybrid-sampling** | Random/distance based over-sampling + under-sampling | 8 papers such as Cateni et al. (2014), Dubey et al. (2014), Díez-Pastor et al. (2015a) | |
| | Cluster-based under-sampling + distance based over-sampling | 8 papers such as Peng et al. (2014), Sáez et al. (2015), Song et al. (2016) | |
| | Other novel methods | Jian et al. (2016) | Over-sampling support-vectors and under-sampling non-support- vectors |

**Table 2**
Summary of articles employing feature selection or extraction methods.

| Article | Method | Detail strategy |
|---|---|---|
| Lusa (2010), Wei et al. (2013b), Lane et al. (2012), Li et al. (2013a), Lima and Pereira (2015) | Filter | Feature ranking based on correlation evaluation (two tailed Student *t*-test, Relief-F, Gini index, Information gain, Chi-Square) |
| Bae and Yoon (2015), Alibeigi et al. (2012), Yang et al. (2016a), Li et al. (2016c), Gong and Huang (2012) | Filter | Feature ranking based on a probability density estimation/ pairwise sequence similarity |
| Beyan and Fisher (2015) Chen and Wasikowski (2008), Maldonado et al. (2014), Trafalis et al. (2014), Casañola-Martin et al. (2016), Al-Ghraibah et al. (2015) | Wrapper | Heuristic search (sequential forward/backward selection, hill-climbing search) |
| Wei et al. (2013a), Li et al. (2016c), Guo et al. (2016) | Wrapper | Stochastic search (random selection, evolutionary method) |
| Dubey et al. (2014) | Embedded | Sparse logistic regression with stability selection |
| Song et al. (2014), Moepya et al. (2014), Vong et al. (2015), Zhang (2016) | Feature extraction | Traditional methods (PCA, NMF, SVD, etc.) |
| Braytee et al. (2016) / Ng et al. (2016) | Novel feature extraction methods | Cost-sensitive PCA and cost-sensitive NMF / Yielded useful feature representation using stacked auto-encoders |

methods, heuristic search was a common choice. Another interesting finding is, that feature selection and extraction were frequently used for solving real-world problem such as disease diagnosis (Casañola-Martin et al., 2016; Dubey et al., 2014; Lusa, 2010; Yang et al., 2016a; Zhang, 2016), textual sentiment analysis (Lane et al., 2012; Zhang et al., 2015a), fraud detection (Li et al., 2013a; Lima and Pereira, 2015 Moepya et al., 2014; Wei et al., 2013b) and other rare events detection problems (Al-Ghraibah et al., 2015; Bae and Yoon, 2015; Gong and Huang, 2012; Guo et al., 2016; Vong et al., 2015), etc.

### 3.1.2. Cost-sensitive learning

By assuming higher costs for the misclassification of minority class samples with respect to majority class samples, cost-sensitive learning can be incorporated both at the data level (e.g. re-sampling and feature selection) and at the algorithmic level (see Section 3.2, López et al., 2012, 2013). The costs are often specified as cost matrices, where $C_{ij}$ represents the misclassification cost of assigning examples belong to class $i$ to class $j$. Given a specific domain, cost matrices can be determined using expert opinion, or in data stream scenarios, they can vary for each record or vary in a dynamic imbalanced status (Ghazikhani et al., 2013b). Compared with re-sampling methods, cost-sensitive learning is more computationally efficient, thus may be more suitable for big data streams. However, this method was used by only 39 of the reviewed papers, which is much less popular than re-sampling methods. There may be two potential reasons, one is that, as stated in Krawczyk et al. (2014), it is difficult to set values in the cost matrix. In most cases, as the misclassification cost is unknown from the data and cannot be given by an expert. Nevertheless, there is an alternative way to address this difficulty, by setting the majority class misclassification cost at 1 while setting the penalty minority class value as equal to the IR (Castro and Braga, 2013; López et al., 2015). Another reason, which may be more reasonable for this observation, is that re-sampling is a common choice in practical for those researchers who are not expert in machine learning. Different from cost-sensitive learning which often needs to modify learning algorithm, re-sampling methods are much easier to be directly implemented in both single and ensemble models. Throughout our study, most application papers employed re-sampling methods instead of cost-sensitive learning.

Three main approaches for dealing with cost-sensitive problems were found in the 39 related papers, as listed in Table 3.

**Table 3**
Summary of articles employing cost-sensitive learning.

| Method | Detail strategy | Articles |
|---|---|---|
| Methods based on training data modification | Modifying the decision thresholds or assigning weights to instance when resampling the training dataset according to the cost decision matrix | Zhang et al. (2016a), Boyu Wang (2016), Zou et al. (2016), Yu et al. (2015), Yu et al. (2016), Voigt et al. (2014), Zhou and Liu (2006) |
| Changing the learning process or learning objective to build a cost-sensitive classifier | Modifying the objective function of SVM/ELM using a weighting strategy | Cheng et al. (2016), Wu et al. (2016), Casañola-Martin et al. (2016), Phoungphol et al. (2012), Maldonado and López (2014) |
| | Tree-building strategies that could minimize misclassification costs | del Río et al. (2014), Krawczyk et al. (2014), Sahin et al. (2013) |
| | Integrating a cost factor into the fuzzy rule-based classification system | López et al. (2015), Vluymans et al. (2015) |
| | Cost sensitive error function on neural network | Oh (2011), Castro and Braga (2013); Ghazikhani et al. (2013b) |
| | Cost-sensitive boosting methods | Sun et al. (2007), Wang et al. (2010) |
| Methods based on Bayes decision theory | Incorporating cost matrix into Bayes based decision boundary | Ali et al. (2016), Datta and Das (2015), Bahnsen et al. (2013), Moepya et al. (2014) |

### 3.2. Classification algorithms for imbalanced learning

Imbalanced learning attempts to build a classification algorithm that can provide a better solution for class imbalance problems than traditional classifiers such as SVM, KNN, decision trees and neural networks. Two methods for solving imbalanced learning problems have been reported in the literature; ensemble methods and algorithmic classifier modifications. In Sections 3.2.1 and 3.2.2, we review the imbalanced data classification algorithms proposed in the past decade. While most of these proposed methods have targeted binary class problems, multi-class issues are commonly seen in many rare event detection domains, such as in machinery fault detection and disease diagnosis. For this reason, multi-class imbalanced learning is discussed as a special issue in Section 3.2.3, which gives a brief introduction to the current solutions.

#### 3.2.1. Ensemble methods

Ensemble-based classifiers, also named multiple classifier systems (Krawczyk and Schaefer, 2013), are known to improve the performance of a single classifier by combining several base classifiers that outperform every independent one (López et al., 2013). Classifier ensembles have become a popular solution method for class imbalance problems. In the 527 reviewed papers, 218 papers proposed novel ensemble models or applied existing ensemble models to solve practical tasks. Galar et al. (2012) thoroughly surveyed imbalanced data learning using ensemble methods, in which ensemble methods were categorized into cost-sensitive ensembles and data pre-processing ensembles. However, as only bagging, boosting and hybrid ensembles (a combination of bagging and boosting) were considered in this study, the field was not fully covered. For example, Sun et al. (2015) and Tian et al. (2011) proposed two ensemble models that trained multiple base classifiers by balancing different datasets created using re-sampling methods without the need for bagging or boosting algorithms.

Note that the training process for the re-sampling based ensemble and bagging can be conducted in parallel, while boosting and some evolutionary based ensemble methods can only be trained using iterative processes. Therefore, in this study, we categorized the ensemble models into two categories, i.e. iterative based ensembles and parallel based ensembles.

*3.2.1.1. Iterative based ensemble.* Boosting is the most common and most effective method for ensemble learning. We found 63 reviewed papers employed boosting in their ensemble framework, most of which were based on the first applicable boosting algorithm, Adaboost, proposed by Freund and Schapire (1996). The benefit of Adaboost is that samples that fail to be assigned to the correct class are given higher weights, which forces the future classifier to focus more on learning these failed classified samples. Adaboost has had several extensions: AdaBoost.M1, AdaBoost.M2, AdaBoost.MR and AdaBoost.MH (Freund and Schapire, 1997; Schapire and Singer, 1999); which were designed for solving multi-class and multi-label problems. Other typical iterative ensemble methods include Gradient Boosting Decision Tree (GBDT) (Friedman, 2001) and some evolutionary algorithm (EA) based ensemble algorithms.

As mentioned in Galar et al. (2012), boosting algorithms are usually combined with cost-sensitive learning and re-sampling techniques. Table 4 lists some of the technical articles that have proposed novel iterative based ensemble algorithms. It can be seen that Adaboost and Adaboost.M2 have been the most popular iterative based ensemble schemes. Most ensemble models have considered cost-sensitive and re-sampling strategies.

*3.2.1.2. Parallel based ensembles.* In this study, parallel based ensembles refer to ensemble models in which each base classifier can be trained in parallel. Parallel based ensemble schemes include bagging, re-sampling based ensembles and feature selection based ensembles. The basic framework for parallel ensemble methods is shown in Fig. 5, in which the dashed boxes and lines represent optional processes. Galar et al. (2012) and López et al. (2013) suggested that bagging and the associated hybridizations with data preprocessing techniques have had good results compared with boosting. Through our research, it was found that parallel ensemble methods dominated two types of papers. First, parallel based ensemble methods were more popular than iterative based ensembles in application-oriented papers (such as in Hao et al., 2014, Wei et al., 2013b, Dai, 2015, etc.), and novel re-sampling methods were often combined with parallel based ensemble schemes (such as in Peng et al., 2014, Sun et al., 2015, Li et al., 2013a, etc.). Since parallel ensembles have time-saving and ease-of-development advantages, they are recommended for solving practical problems.

*3.2.1.3. Base classifier choice in ensemble models.* When implementing iterative or parallel ensemble methods, a base classifier is needed, which can be any of the classical models such as SVM and neural networks. Fig. 6 summarizes the main base classifiers selected by proposed ensemble learning algorithms. Note that some papers examined multiple base classifiers. We found that SVM, NN (neural network), NB (naïve Bayes), rule-based classifiers, and decision tree based classifiers (includes C4.5, CART, random forest and other novel tree classifiers) have been the most selected in the literature. Sun et al. (2009) summarized the difficulties of some base classifiers when learning from imbalanced data and pointed out

**Table 4**
Representatives of reviewed iterative ensemble methods.

| Algorithms and reference | Ensemble scheme | Combined strategies |
|---|---|---|
| AdaC1, AdaC2, and AdaC3 (Sun et al., 2007), BoostingSVM-IB (Zięba et al., 2014), Boosting-SVM (Wang and Japkowicz, 2010), Adaboost.NC (Wang et al., 2010) | Adaboost (slightly modified re-weighting strategy) | Cost-sensitive |
| RUSBoost (Seiffert et al., 2010) /GESuperPBoost (García-Pedrajas and García-Osorio, 2013) | AdaBoost.M2 / Adaboost (the weighted error is modified) | RUS(Under-sampling) |
| EUSBoost (Galar et al., 2013), RAMOBoost (Chen et al., 2010), BNU-SVMs (Bao et al., 2016b) | AdaBoost.M2 or Adaboost | Novel under-sampling |
| MSMOTEBoost (Hu et al., 2009; Ren et al., 2016a) | Adaboost | Novel over-sampling |
| EasyEnsemble and BalanceCascade (Liu et al. 2009; Hassan and Abraham, 2016) | Adaboost | Bagging, RUS(Under-sampling) |
| BSIA (Zięba and Tomczak, 2015) | Adaboost | Under-sampling, cost-sensitive |
| Prusa et al. (2016) and Guo et al. (2016) | Adaboost, Adaboost.M1 | Under-sampling, Feature selection |
| Boyu Wang (2016), Ditzler and Polikar (2013), Wang et al. (2013), Dal Pozzolo et al. (2015) | Adaboost | Online boosting, re-sampling |
| Li et al. (2014) | Adaboost | Weighted ELM |
| RB-Boost (Díez-Pastor et al., 2015a), ESM-UP-DW (Bae and Yoon, 2015) | Adaboost or Adaboost.M2 | Hybrid-sampling |
| RankCost (Wan et al., 2014; Lusa, 2016) | GBDT | Cost sensitive |
| Can-CSC-GBE (Ali et al., 2016) | Gentleboost | Cost-sensitive |
| DyS (Lin et al., 2013a) | Novel iterative learning scheme | Dynamic sampling |
| M-AdaBoost cascade decision tree (Yi, 2010) | Novel cascaded structure based on Adaboost | |
| Krawczyk et al. (2014), Krawczyk et al. (2016) and Folino et al. (2016) | Evolutionary algorithm based ensemble | Cost sensitive, Feature selection |
| LEAT (Park and Ghosh, 2014) | Novel boosting mechanism | |

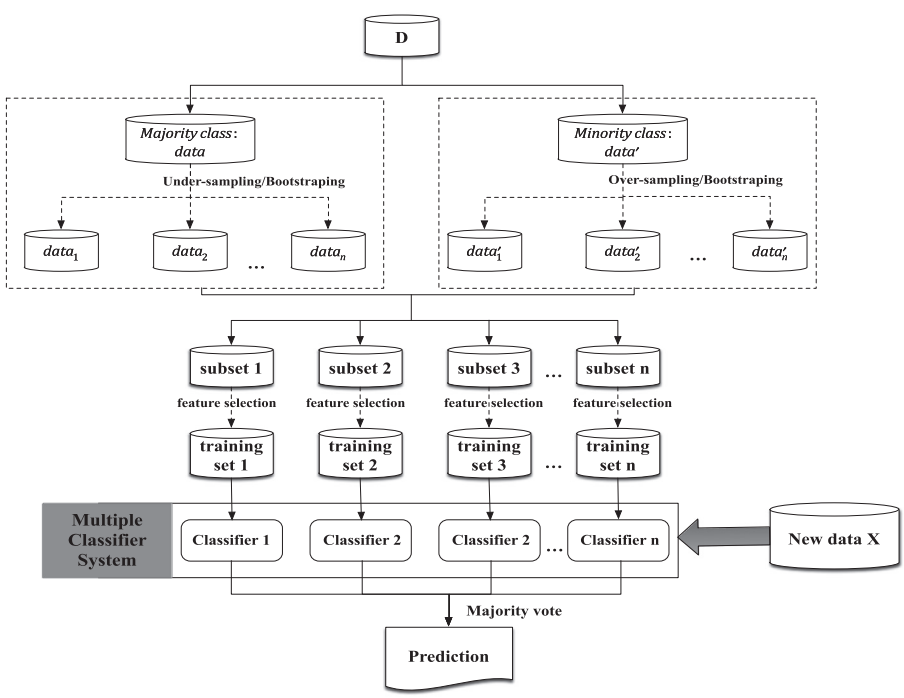


**Fig. 5.** Parallel ensemble framework (dashed frames and lines stand for optional operations).

that there were dozens of classifiers for machine learning, each with their own strengths and weaknesses. Users need to choose a proper base classifier grounded in reality. For example, SVM is robust and precise, but can be sensitive to missing values and difficult to train for large scale data. In contrast, decision trees are good at handling missing value cases but can fail to model small size data (Li et al., 2016c).

*3.2.2. Algorithmic classifier modifications*

Improving the learning ability of existing classification algorithms to improve the classification performance for imbalanced data is another main imbalanced learning research direction. Over 160 novel improved classifiers have been proposed in the past decades in the class imbalanced learning and rare event detection fields. SVM, decision tree, NN, KNN, ELM, rule-based classifiers, and

**Table 5**
Representatives of reviewed algorithmic classifier modifications.

| Strategy | Detailed description | Representative articles |
|---|---|---|
| Kernel and activation function transformation method (Mostly based on SVM, ELM, and NN) | Enhances the discriminatory power of the classifiers using kernel transformation to separate two classes close to the boundary as far from each other as possible<br>Designing mixed kernel/basis functions to increase the separability of the original training space | Logistic regression: Maalouf and Trafalis (2011); SVM: Zhang et al. (2014), Maratea et al. (2014), Zhao et al. (2011); ELM: Gao et al. (2016), Wang et al. (2016b); NN: Raj et al. (2016), da Silva and Adeodato (2011)<br>SVM: Chen et al. (2012); NN: Pérez-Godoy et al. (2010); ELM: Wu et al. (2016) |
| Objective transformation method | Converting the training goal to a well-deigned objective function that more severely penalize errors in the minority examples<br>Multi-objective optimization | Cost-sensitive learning: See Section 3.1.2; Weighted KNN: Kim et al. (2016); SVM: Duan et al. (2016b);<br>SVM: Shao et al. (2014), Datta and Das (2015), Xu et al. (2015); Bayesian NN: Hong et al. (2007), Lan et al. (2009); KNN: Ando (2015), Huang et al. (2006b) |
| Fuzzy based method | Fuzzy rule-based classifiers (FRBCs) and decision trees to extract classification rules from the imbalanced data<br>Other fuzzy based classifiers | Alshomrani et al. (2015), Fernández et al. (2010b), Ramentol et al. (2015), Vluymans et al. (2015), Bagherpour et al. (2016)<br>SVM: Cheng and Liu (2015); KNN: Liao (2008) NN: Gao et al. (2014), Tan et al. (2015b) |
| Clustering algorithm | Unsupervised learning | Liang et al. (2012), Vigneron and Chen (2015), Fan et al. (2016), Zhang (2016) |
| Task decomposition strategies | A hierarchical decomposition strategy to globally and locally learn from the imbalanced data | Naïve Bayes and NN: Lerner et al. (2007); Rule based classifier: Napierała and Stefanowski (2015), Jacques et al. (2015); KNN: Garcia-Pedrajas et al. (2015), Beyan and Fisher (2015) ELM: Xiao et al. (2016), Mao et al. (2016) |

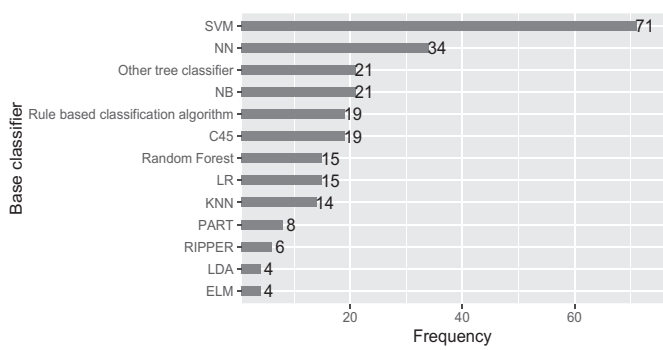


**Fig. 6.** Statistics for base classifiers. (NN: Neural Network, NB: Naïve Bayes, LR: Logistic Regression, PART: Projective ART (Cao and Wu, 2002), RIPPER: Repeated Incremental Pruning to Produce Error Reduction (Natwichai et al., 2005), LDA: Linear Discriminant Analysis, ELM: Extreme learning machine (Huang et al., 2006a).

naïve Bayes modifications were the most used at 54, 33, 24, 15, 13, 11, and 9 papers, respectively. Some common techniques for improving these 6 classifiers are summarized in Table 5.

### 3.2.3. Multi-class imbalanced learning

Multi-class learning has been seen as a difficult task for classification algorithms as multi-class classification may have a significantly lower performance than binary cases. This issue becomes more complex when facing imbalanced data, as the boundaries between the classes can severely overlap (Fernández et al., 2013). Multi-class imbalanced learning has attracted significant attention in recent years. A total of 32 of the reviewed papers generalized binary class imbalanced solutions into multi-class cases. Two generalization methods appeared to be the most used; a one-versus-one approach (OVO) and a one-versus-all approach (OVA), both of which are based on decomposition techniques. The OVO and OVA decomposition schemes are shown in Fig. 7 (Zhou, 2016), in which $C_i$ represents all examples labeled $i$, and $f_j$ is the hypothesis generated by classifier j.

Fernández et al. (2013) studied the OVO and OVA decompositions, and ad-hoc learning algorithms, which are natural for addressing multiple class learning problems. The experimental results showed that OVO outperforms OVA. However, there is no significant differences between decomposition methods and standard ad-hoc learning algorithms. Similar conclusions can be found in Wang and Yao (2012). They believe that it is unnecessary to use class decomposition, while learning from the entire data set directly is sufficient for multi-class imbalance classification. They concluded that it is unwise to integrate class decomposition with class imbalance techniques without considering the class distribution globally.

Although the above papers suggested that OVA is not the best available tool, it is still popular as it requires less decomposition thus is time efficient. Relative to OVA, OVO was less popular. A summary of OVO, OVA, ad-hoc related articles are shown in Table 6.

## 3.4. Model evaluation in the presence of rare classes

Model selection and model evaluation are two crucial processes in machine learning. Performance measures, therefore, are key indicators for both evaluating the effectiveness and guiding the learning of a classifier. Accuracy is the most commonly used evaluation metric for classification. However, under imbalanced scenarios, accuracy may not be a good choice because of the bias toward the majority class. Performance metrics adapted into imbalanced data problems, such as Receiver Operating Characteristics (ROC), G-Mean($GM$), and F-measure($F_m$), are less likely to suffer from imbalanced distributions as they take class distribution into account. Since these measures are widely used in imbalanced learning field and their detailed formulas can be found in most imbalanced learning related papers (such as Branco et al., 2016 and López et al., 2013), we only introduce these metrics in Appendix (see Supplementary Material). There are also some works focused on proposing novel evaluation metrics for imbalanced data, such as Adjusted F-measure (Maratea et al., 2014) and a probabilistic thresholding method (Su and Hsiao, 2007).

The most frequently used metrics include Accuracy, AUC/ROC, F-Measure, G-mean, Precision, Sensitivity, Specificity, Balanced accuracy, and Matthews Correlation Coefficient (MCC). Two papers

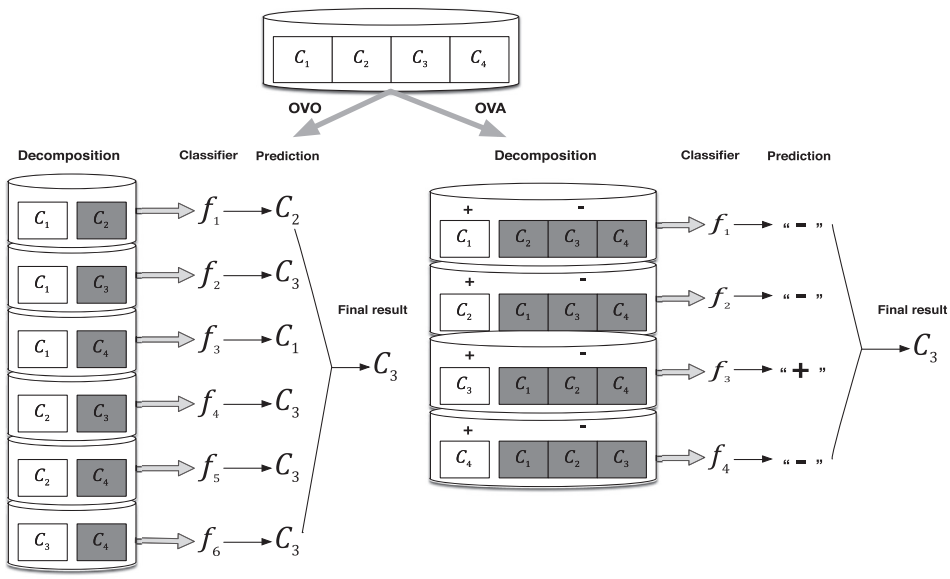


**Fig. 7.** OVO and OVA decomposition schemes. (Note: In the OVA scheme, one class is treated as a positive class and denoted "+", while the other three classes are merged as a negative class and denoted as "-". Based on the predictive labels for the four classifiers on the right side, it is clear that final result is $C_3$).

**Table 6**
Representatives of reviewed multi-class imbalanced learning algorithms.

| Strategy | Detailed description | Representative articles |
|---|---|---|
| OVA | Re-sampling techniques with OVA<br>Combining OVA with ensemble learning | Liao (2008), Zhao et al. (2008), Chen et al. (2006)<br>Zhang et al. (2015d), Yang et al. (2016a) |
| OVO | Re-sampling with OVO<br>Combining OVO with ensemble learning | Fernández et al. (2010a)<br>Zhang et al. (2016b), Cerf et al. (2013) |
| Ad-hoc | Directly generalizing learning algorithms in to multi-class cases | SVM: Duan et al. (2016a), Shao et al. (2014); ELM: Yang et al. (2016a), Yu et al. (2016), Mirza et al. (2015a), Mirza et al. (2013) |

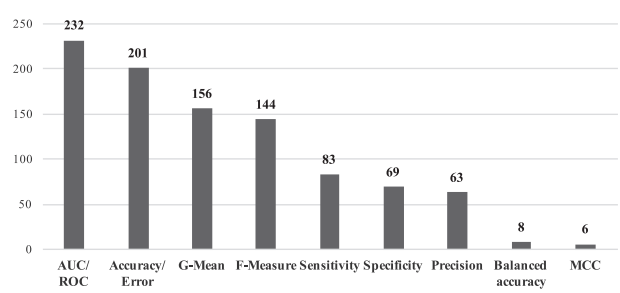


**Fig. 8.** The number papers using the evaluation metrics.

investigated the effectiveness of these metrics for imbalanced learning and made some recommendations (Gu et al., 2009; Jeni et al., 2013). A usage summary for all the metrics is given in Fig. 8. Note that although accuracy based metrics have been proven to be biased towards the majority class, these are still frequently used in research because they are the most general and intuitionistic metrics for classification tasks. AUC, the G-mean, and the F-measure have also been often utilized as evaluation measures for both model comparisons and model selections.

It should be noted that AUC/ROC has been questioned by Hand (2009), who argued that ROC depends on cutoffs the model produces and the cutoffs are related to the misclassification costs when only optimal thresholds are considered, hence the ROC of a model is incoherent. However, there are also some counter arguments about this interpretation (Ferri et al., 2011). Generally, AUC/ROC is recognized as a useful measure of ranking performance.

Metrics mentioned above are only applicable to binary classification problems. A natural way to extend these metrics to multi-class cases is to employ decomposition methods (the OVA and OVA schemes described in 3.2.3) and take the average of each pair-wise metric (Cerf et al., 2013). MAUC (Hand and Till, 2001) is an example of a mean AUC and in Guo et al. (2016) and Li et al. (2016c), a derivation of AUC was used. Another AUC based multi-class metric, Volume Under ROC (VUC) was employed in Phoungphol et al. (2012) and an extension of the G-mean was adapted by Sun et al. (2006). For the F-measure, two types of averages, micro-averages and macro-averages were introduced in Phoungphol et al. (2012).

## 4. Imbalanced data classification application domains

There is currently a great deal of interest in utilizing automated methods—particularly data mining and machine learning methods—to analyze the enormous amount of data that is routinely being collected. An important class of problems involves predicting future events based on past events. Event prediction often involves predicting rare events (Weiss and Hirsh, 2000). Rare events are events that occur with low frequency but may cause far-reaching impact and disrupt the society (King and Zeng, 2001). Rare events lurk in many shapes, including natural disaster (e.g. earthquakes, solar flares, tornado), anthropogenic hazards (e.g. financial fraud, industrial accidents, violent conflict) and diseases. Since such data are usually imbalanced, many efforts have been directed to build rare events detection systems with the help of imbalanced learning methods. Among the literature, we find 162 articles are application-oriented, ranging from management to engineering. In Section 4.1, we develop a taxonomy scheme to categorize these 162 articles into 13 application fields of rare events detection. In Sections 4.2–4.14, each category and its sub-categories are described and a few examples are given.

**Table 7**
Application domain categories.

| Application domains | No. of papers |
|---|---|
| (1) Chemical, biomedical engineering | 47 |
| (2) Financial management | 37 |
| (3) Information technology | 24 |
| (4) Energy management | 8 |
| (5) Security management | 7 |
| (6) Electronics and communications | 6 |
| (7) Infrastructure and industrial manufacturing | 9 |
| (8) Business management | 7 |
| (9) Emergency management | 4 |
| (10) Environmental management | 5 |
| (11) Policy, social and education | 4 |
| (12) Agriculture and horticulture | 1 |
| (13) Other areas and non-specific areas | 3 |

The main purpose of this section is to show researchers from different areas how imbalanced learning has been applied to detect rare patterns/events in their research domains. As learning techniques have already been presented in Section 3, we try to omit technical details in this section to avoid overlapping and focus more on describing specific problems and corresponding data collection and feature engineering processes in this section. Also, we found some interesting observations about which methods tend to be popularly used in different domains through our study, which are concluded in Section 6.

### 4.1. Taxonomy of application domains

Excluding popular benchmarks such as UCI and KEEL, we categorize the real-world applications into 13 categories. Each category contains several topics. We divided managerial applications into six categories: finance, energy, security, emergency, environment, and business management. The remaining categories involve engineering and humanity fields and are divided into: chemical, biological and medical engineering; information technology; electronics and communications; infrastructure and industrial manufacturing; policy, social and education; agriculture and horticulture; other areas. Table 7 shows the taxonomy and frequency distribution of the papers. Note that some applications may be interdisciplinary, we simply categorized them into the groups best describing their domain.

### 4.2. Chemical and biomedical engineering

Biomedical engineering applies engineering principles and design concepts to medicine and biology for healthcare purposes (e.g. diagnosis, monitoring and therapy), while chemical engineering tries to convert chemical materials or cells into useful forms and products. Both biomedical and chemical engineering employ physical and life sciences together with applied mathematics, computer science and economics. The application often involves building a decision support system to detect and predict abnormal structures in chemical processes and biomedical activities. These include disease diagnoses, disease early warning, protein detection, chemical repulsion and drug resistance. The related research problems and corresponding references are shown in Table 8.

The most researched topics related to imbalanced chemical and biomedical data classification problems were protein detection, gene expression prediction, and diseases diagnoses. Protein data sets are usually imbalanced data sets, and protein detection tries to identify protein structures and functions based on their sequence expressions (Dai, 2015). Feature extraction is essential in protein detection tasks to convert the non-numerical attributes in the sequences to numerical attributes (Vani and Sravani, 2014; Wasikowski and Chen, 2010). Similar properties can be found in gene expression identification (Yu et al., 2012) and DNA identification (Song et al., 2014). Image data are also utilized frequently to analyze abnormal biomedical events: Bria et al. (2012) used 198 images to detect microcalcifications on digital mammograms; Lerner et al. (2007) analyzed abnormal genes using FISH signal images; Bae and Yoon (2015) focused on finding the locations and sizes of polyps in endoscopic or colonoscopic images.

### 4.3. Financial management

Financial management is a sub-area of business management. Our review found 37 papers have tackled financial problems; we thus separated financial management from business management. Financial management is an activity of management which is concerned with the planning, procuring and controlling of the firm's financial resources. Sanz et al. (2015) tested imbalanced data classification methods in 11 financial problems including stock market prediction, credit card/loans approval application system and fraud detection. Other papers and their application setting are given in Table 9.

Most papers in this category were focused on financial fraud detection, which included e-payment fraud, credit and plastic card fraud, fraud in firm's activity, insurance fraud, and fraudulent company financial statements. Krivko (2010) highlighted the several challenges in building an effective fraud detection system; large volumes of daily transaction records, low frequency fraud occurrence and information delay. Detecting fraud events is a typical imbalanced learning problem as transaction records are highly skewed. In general, the dataset used to train a fraud detection system includes customer profiles and transaction records (type of transaction, date, location, amount, etc.). Kim et al. (2012) described some representative features of delinquency information used in credit card and loan fraud detection. Other interesting research topics also have been studied. Based on San Diego real estate information, Gong and Huang (2012) collected 37 variables on a property's status and whether it was refinanced to predict property refinancing. Alfaro et al. (2008) and Zhou (2013) used various features as the factors to judging corporate life cycle stages or even to predict corporate bankruptcy: dividend payout ratio, sales growth, capital expenditure, the firm age and other firm profiles such as recorded liabilities, assets, sales, legal structures and tax information. Other sources of data like network behavior information, social network information and other web information were also utilized in the literature (Abeysinghe et al., 2016).

### 4.4. Information technology

Information technology (IT) is the application of computers to store, retrieve, transmit and manipulate data. With the explosive growth in web data, detecting interesting events from information devices and platforms is crucial to business decision-making and strategy formation. Traditionally, software defect detection, network intrusion detection and other anomaly detection methods are implemented under imbalanced scenarios. The collected articles in this domain were divided into three parts according to the research subject, as shown in Table 10.

Predicting software defects and quality were the two primary research topics in software engineering. Module attributes are commonly used to predict defects or evaluate the quality of software. On the other hand, network intrusion detection often needs to make predictions online, which leads to an online imbalanced learning problem, detailed techniques was introduced in Wang et al. (2013). With the rapid development of internet technology, web data has become an important resource to analyze customer preferences. Vorobeva (2016) used web-post to identify web author, and sentiment analysis has become a popular

**Table 8**
Applications in chemical and biomedical engineering fields.

| Research topic | Detailed application | The literature |
|---|---|---|
| Diseases diagnosis | Alzheimer's disease | Dubey et al. (2014) |
| | Transient ischemic attack prediction | Cheng and Hu (2009) |
| | Lung nodule detection | Peng et al. (2014) |
| | Cancer diagnosis | 11 papers such as Krawczyk et al. (2016), Yang et al. (2016a), Lusa (2010), Yu et al. (2012) |
| | Coating on tongue | Li et al. (2013b) |
| | Diabetes | Vo and Won (2007), Oh et al. (2011) |
| | Hepatitis virus detection | Oh et al. (2011), Richardson and Lidbury (2013) |
| | Traumatic events detection | Niehaus et al. (2014) |
| | Polyp detection | Bae and Yoon (2015) |
| | pediatric acute conditions | Wilk et al. (2016) |
| | Parkinson's Disease | Yeh et al. (2016) |
| Protein/DNA/Gene related identifications | Protein/DNA identification | 7 papers such as Herndon and Caragea (2016), Song et al. (2014), Vani and Sravani (2014), Zou et al. (2016) |
| | Protein-protein interaction | Bao et al. (2016a), Li et al. (2009), Zhang et al. (2012) |
| | Gene expression prediction | Yu et al. (2012), Blagus and Lusa (2013) |
| | Genetic abnormalities | Lerner et al. (2007) |
| Monitoring and therapy | Length of stay for appendectomy patients | Cheng and Hu (2009) |
| | Microcalcification detection | Bria et al. (2012) |
| | Liveability prediction | Zięba et al. (2014), Li et al. (2015) |
| | Risk scoring for the prediction of acute cardiac complications | Liu et al. (2014) |
| | Tuberculosis early warning | Li et al. (2016b) |
| | Elucidate therapeutic effects in uremia patients | Chen (2016) |
| Chemical engineering | Small molecules detection | Hao et al. (2014) |
| | Drug resistance | Raposo et al. (2016) |
| | Identifying potential new drugs | de Souza et al. (2016) |
| | Biological absorption, distribution, metabolism, and excretion prediction | Casañola-Martin et al. (2016), |
| Biology | Animal behavioral task and chronic neural recording | Vigneron and Chen (2015) |

**Table 9**
Applications related to financial management.

| Research topic | Detail application | The literature |
|---|---|---|
| Fraud detection | Online banking/web payment system fraud | Zhang et al. (2008), Wei et al. (2013b), Lima and Pereira (2015) |
| | Credit/plastic card fraud detection | 13 papers such as Zakaryazad and Duman (2016), Sanz et al. (2015), Sahin et al. (2013) |
| | Occupational fraud detection | Mardani and Shahriari (2013) |
| | Insurance fraud detection | 6 papers such as Hassan and Abraham (2016), Pan et al. (2011), Kirlidog and Asuk (2012) |
| | Fraudulent financial statement detection(Firms) | 6 papers such as Li and Wong (2015), Moepya et al. (2014), Pai et al. (2011) |
| | Tax declaration | Zhang et al. (2009) |
| Property refinance prediction | Predict whether or not a particular property would refinance | Gong and Huang (2012) |
| Loan default prediction | Predict whether a lender would repay in time | Abeysinghe et al. (2016), Brown and Mues (2012), Sanz et al. (2015) |
| Corporate bankruptcy prediction | Pre-warning of whether a corporate will fall into a decline stage or even bankruptcy | Lin et al. (2013b), Alfaro et al. (2008), Zhou (2013) |
| Credit card approval | Whether to approve a new applicant | Li and Wong (2015), Sanz et al. (2015) |

**Table 10**
Information technology applications.

| Research subject | Detail application | The literature |
|---|---|---|
| Software | Software defect prediction | Wang and Yao (2013), Rodriguez et al. (2014), Tan et al. (2015a) |
| | Software quality evaluation | Drown et al. (2009) |
| | Automated identification of high impact bug reports | Yang et al. (2016b) |
| Network | Network intrusion detection | 8 papers such as Folino et al. (2016), Hajian et al. (2011), Engen et al. (2008) |
| | Peer-to-peer traffic classification | Zhong et al. (2013) |
| Web | Data disclosure prediction | Zhang et al. (2015c) |
| | Web author identification | Vorobeva (2016) |
| | Web service Qos prediction | Xiong et al. (2014) |
| | Click fraud in online advertising | Taneja et al. (2015) |
| | Fake website detection | Abbasi and Chen (2009) |
| | Sentiment analysis | 6 papers such as Prusa et al. (2016), Lane et al. (2012), Zhang et al. (2015a) |

topic which processes and analyzes users' preferences from User-Generated Content (UGC) on the web. Since users' opinions on a specific topic tend to be consistent, UGC data for sentiment analysis is usually imbalanced. As sentiment analysis and other web data mining models are often built from heterogeneous data sources such as text, image and numeric data, feature engineering techniques are needed to generate the features before the imbalanced learning models can be designed. For example, word embedding (Mikolov et al., 2013) is an efficient technique to build word representations for text, and Convolutional Neural Networks (CNN, or ConvNet) are popular for generating features from raw images (Razavian et al., 2014).

### 4.5. Energy management

Energy management includes the planning and operation of energy production and energy consumption units. There were 8 papers related to this area. Guo et al. (2016) and Li et al. (2016c) used oil well logging data to recognize the oil-bearing formations of each layer in the well. Xu et al. (2007a, b) focused on power distribution outage identification to enhance the availability and reliability of power distribution systems. Historical power distribution outage data as well as environmental attributes were utilized to extract fault patterns caused by trees, animals and lightning. A condition assessment model was built by Ashkezari et al. (2013) to evaluate the healthiness(fitness) level of power transformers. The model was trained by dissolved gas analysis and insulating oil test data. Qing (2015) focused on predicting the inventory material consumption demand of an electric power system. They found project-based demand and operational/maintenance-based demand in electric power systems have skewed frequencies; therefore, building an imbalanced learning model able to forecast possible material consumption demand can have better results than traditional machine learning models. Fraud detection techniques were also applied in the energy field. In particular, electricity fraud, which is the dishonest or illegal use of electricity equipment or service with the intention to avoid billing charges (Fabris et al., 2009; Nagi et al., 2008). Since electricity customer consumption data is made up of time series records, these attributes need to be treated in a special manner to extract all the relevant information (Fabris et al., 2009).

### 4.6. Security management

To implement effective controls, organizations use security management procedures such as potential risk detection, risk assessment and risk analysis to identify threats, crimes, and other anomalies. Insider threats such as infringing intellectual property and malicious sabotage are crucial events that need to be detected by a security department. Azaria et al. (2014) analyzed the behavior of insider threat. Experiments using Amazon Mechanical Turk (AMT) were conducted to distinguish normal behaviors and of those who intend to leak private data from an organization. Another common application in security management was detecting threats and unusual events from surveillance video (Mandadi and Sethi, 2013; Wang et al., 2016a; Wang et al., 2015b; Wen et al., 2015; Xu et al., 2016). Automatic event detection systems based on surveillance video typically contained feature extraction and pattern classification components (Xu et al., 2016). With regards to feature extraction, approaches such as discrete optical flow descriptors, trajectory based approaches (Xu et al., 2014) and sparse spatio-temporal (ST) corner descriptors were some of widely-used methods to extract features from local image regions and video clips (Mandadi and Sethi, 2013). For the pattern classification phase, imbalanced data classification algorithms can be used to recognize abnormal actions and events.

### 4.7. Electronics and communications

Five papers were related to electronics and telecommunications in our study, four of which focused on detecting telecommunications fraud. In brief, telecommunications fraud can be simply defined as any activity in which a telecommunications service is obtained without any payment intention (Hilas and Mastorocostas, 2008). Hilas and Mastorocostas (2008) examined several telecommunication fraud categories such as technical fraud, contractual fraud, hacking fraud and procedural fraud. Similarly, Farvaresh and Sepehri (2011) identified subscription fraud, dial through fraud, free phone fraud, premium rate service fraud, handset theft, and roaming fraud, and sought to detect residential and commercial subscription fraud based on call detail recording and bill data. Other studies such as Olszewski (2012) and Subudhi and Panigrahi (2015) distinguish normal and fraudulent behavior based on user profiles.

Other papers in this category were Kwak et al. (2015) and Tan et al. (2015b) both of which examined the detection of circuitry defects or other anomalies in a wafer of electrical and electronic devices. Tan et al. (2015b) trained a machine learning model on an encrypted dataset generated from three main semiconductor manufacturing operational processes (etests, sort and class test).

### 4.8. Infrastructure and industrial manufacturing

In this category, eight papers applied imbalanced learning methods to solve industrial manufacturing problems. Cateni et al. (2014) used resampling method for two metal industry problems. The first problem concerned the detection of defects in an automatic inspection system of the product surface during production. The other industrial application from the steel-making field was designed to identify nozzles to determine final product quality. Sun et al. (2010) focused on a detection process for variations in the nano-CMOS circuits used widely in manufacturing. The proposed method was tested on six instrument indicators and realized with a 45 nm CMOS. Liao (2008) proposed a multi-class imbalanced data classification algorithm to identify the different types of weld flaws that might be unequally distributed. In their experiments, each weld flaw sample had 12 features describing the shape, size, location, and intensity information, which were extracted from a radiographic image. The goal of their study was to classify weld flaws to identify a lack of fusion, a lack of penetration, gas holes, porosities or cracks. Tajik et al. (2015) proposed a fault detection system for industrial gas turbines. Applying imbalanced learning to machinery fault diagnoses has been studied recently: Duan et al. (2016a) and Mao et al. (2017) classified multiple faults that could occur in roller bearings; Jin et al. (2014) and Zhang et al. (2015b) constructed several features to represent different health conditions of machines in order to detect potential machinery faults.

There is also one paper related to building modeling. Xin et al. (2011) suggested that detecting building foot-points from LIDAR data was the foundation and one of the difficulties in building modeling and edge detection applications. They therefore attempted to detect building points from a non-ground points dataset, which consisted of two unbalanced datasets generated from a built-up area (with dense buildings and small tress) and a rural area (with dense trees and low houses).

### 4.9. Business management

Business management is a broad concept that includes planning, organizing, staffing, leading, and controlling an organization to accomplish a goal or target. As financial management was introduced in Section 4.3, only 7 papers that focused on other busi-

ness functions are discussed in this category, most of which were related to Customer Relationship Management (CRM). Data mining is an essential part of CRM to analyze large data streams and gain insight into customer behavior, needs and preferences (Lessmann and Voß, 2009).

Sultana (2012) used customer data from an insurance company to identify the potential customers that preferred to buy caravan insurance. The features of interest were socio-demographic variables derived from the customer's ZIP area code, and variables regarding the ownership of other insurance policies. Chen et al. (2012), Verbeke et al. (2012), Wu and Meng (2016), and Yi (2010) chose purchase time, purchase amount, rebates, buyer's credit rating, payment integral and demographic information as features to detect customer churning behavior. Chang and Chang (2012) applied an imbalanced learning model to monitor online auctions, in which attributes such as density of ratings, time information and historical records were used to detect significant abnormal and fraudulent behaviors. Bogina et al. (2016) leveraged the temporal features of both the session and the items clicked in a session to predict whether it ends up with a purchase.

### 4.10. Emergency management

To our surprise, few reviewed papers casted attentions on predicting emergency events. An emergency is a situation that poses an immediate risk to health, life, property or the environment (Anderson and Adey, 2012). Emergency events are typically rare events considering their infrequency. Due to their sudden destructiveness, predicting emergency events is a valuable yet difficult research topic. Existing research in emergency event detection has focused on natural disasters. Maalouf and Trafalis (2011), Maalouf and Siddiqi (2014), and Trafalis et al. (2014) built imbalanced learning models to forecast tornados. The tornado dataset had 83 attributes, including radar-derived velocity parameters that described aspects of the mesocyclone, monthly attributes, features that described the pre-storm environment and the predisposition of the atmosphere to explosively lift air over specific heights. Kim et al. (2016) applied their imbalanced learning model to detect several emergencies such as earthquake, fire, flood, landslide, nuclear event, and volcano using text documents collected from the Hungarian National Association of Radio Distress-Signaling and Infocommunications (RSOE, monitors extraordinary risk events that occur all over the world, 24 h per day).

### 4.11. Environmental management

Environmental resource management is the management of the interaction and impact of human societies on the environment. Vong et al. (2015) suggested that air pollution index forecasts were a time series problem. In their study, an online sequential learning method was used to predict the $PM_{10}$ level (Good, Moderate, and Severe). Air pollution data collected by the Macau government meteorological center (including atmospheric pressure, temperature, mean relative humidity, wind speed, etc.) was selected as the case study. Topouzelis (2008) focused on ocean oil spills, which can seriously affecting the marine ecosystem. The amount of pollutant discharges and the associated effects on the marine environment was employed to evaluate sea water quality. Similarly, in order to monitor oil spill events, Brekke and Solberg (2008) used Synthetic Aperture Radar images (SAR) to distinguish oil spill from other natural phenomena.

Other two papers focused on predicting pollution threshold exceedances. Lu and Wang (2008) employed a cost sensitive algorithm with SVM to predict ozone threshold exceedances (pollutant day), and Tsai et al. (2009) applied cost sensitive neural network methods to forecast the ozone in an episode day. Both studies proved that cost-sensitive algorithms can effectively address imbalanced data issues and obtain a better forecast of rare samples (pollution days) in environmental applications. However, the application of re-sampling or ensembles to environmental management has not yet been researched.

### 4.12. Policy, social and education

Three key concepts related to public utilities were integrated in this category, but only four papers focused on social and educational issues. Márquez-Vera et al. (2013) posited that detecting student failure was an effective way to better understand why so many youths fail to complete their school studies. This was a difficult task as there are many possible factors that could affect school failure; further, a majority of students pass. Therefore, this failure was recognized as a prediction problem with high dimensional, imbalanced data. In their study, 77 attributes (e.g. socioeconomic factors; personal, social family and school factors; previous and current marks) were selected to build the imbalanced data classification model to predict whether a student would pass or fail high school. Other interesting studies include: Huang et al. (2016) analyzed video data to predict crowd counting; Ren et al. (2016b) designed a comprehensive feature engineering process to predict potential red light running in real time, features they used include occupancy time, time gap, used yellow time, vehicle passing, etc. Sensor data was generated from wearable devices by Gao et al. (2016) to monitor the occurrence of fall accidents.

### 4.13. Agriculture and horticulture

Agriculture and horticulture is an important area in science. However, we found only one paper in this group. D'Este et al. (2014) tackled a shellfish farm closure problem by predicting the desired shortest possible closure time based on water quality information. The dataset used was 18,692 manual water samples taken by the Tasmanian Shellfish Quality Assurance Program from 38 growing zones in Australia.

### 4.14. Other areas and non-specific applications

Applications that did not fit any of the 12 categories were assigned to this category. In detail, we found two papers related to astronomical research. For instance, Voigt et al. (2014) studied gamma-ray astronomy detection problem, where hadron observations are 100 to 1000 times more common than gamma events. Data from the MAGIC experiment in astronomy was collected to choose an optimal threshold for signal-background-separation. Al-Ghraibah et al. (2015) attempted to detect solar flares by predicting flare activity from quantitative measures of the solar magnetic field. Finally, Vajda and Fink (2010) and Alsulaiman et al. (2012) proposed a handwritten recognition system under an imbalanced scenario to identify verification.

## 5. Future research directions of imbalanced learning

In this section, we propose possible research directions of imbalanced learning based on our survey. In particular, imbalanced techniques we think still need to be considered are proposed in Section 5.1. Some application domains that imbalanced data are frequently observed but not well-studied are pointed out in Section 5.2.

### 5.1. At the technical level

#### 5.1.1. Diversity within ensembles

As a good way to improve the classification performance of weak learners, ensemble based algorithms have been employed to solve many imbalanced learning tasks.

Wang and Yao (2009) maintained that the performance of an ensemble model was decided by the accuracies of the individual classifiers and the diversity between all classifiers. Diversity is the degree to which classifiers make different decisions on one problem. Diversity allows voted accuracy to be greater than that of single classifier. They demonstrated how diversity affected classification performance, especially on minority classes. Their empirical studies have shown that a larger diversity results in a better recall for the minority but harms the majority classes as it enhances the probability of classifying the examples as minority when the accuracy is not high enough. In addition, multi-classes are more flexible and beneficial when increasing diversity within an ensemble model. A similar study can be found in Błaszczyński and Lango (2016).

Several previous works have taken diversity into consideration when building the ensemble model, such as Díez-Pastor et al. (2015b), Krawczyk and Schaefer (2013),and Lin et al. (2013a), in which diversity measures or evolutionary methods were employed to prune the classifiers in an ensemble model to maintain the diversity. However, diversity problems still need to be studied carefully as most existing applications tend to first learn accurate base classifiers which are then integrated into the ensemble. Wang and Yao (2009) suggested that the status in an ensemble model with medium accuracy and medium diversity could lead to better performance, but the trade-off between accuracy and diversity remains unclear. Further, related to this, while pruning classifiers can be powerful in increasing ensemble diversity and avoiding overfitting, many base classifiers still need to be trained and evaluated before the pruning process, which is time-consuming. Building an ensemble model that integrates diversified and precise weak learners more efficiently needs to be addressed in future studies.

#### 5.1.2. Adaptive learning

Hundreds of algorithms have been proposed to deal with imbalanced data classification problems, and they are shown to outperform others in some dimension. However, from the technical papers, we found no specific algorithm that was superior in all benchmark tests. Most proposed algorithms treated all imbalanced data consistently and handled it using a versatile algorithm. Yet, as imbalanced data has variations in the imbalanced ratio, the number of features and the number of classes, the classifier performances when learning from different types of datasets are different. This uncertainty in a learning model becomes more obvious when building ensemble models. Li et al. (2016c) suggested that using a specific ensemble classifier to tackle all kinds of imbalanced data was inefficient. The learning quality of a model can be affected by the way the training samples are selected, the options of the base classifiers and the final ensemble rules. While building a unified ensemble framework has been well-studied in the past decade, each component within an ensemble framework is often decided by the users. This raises another question as to how to adaptively choose a detailed algorithm to fit each component in an ensemble framework for different types of imbalanced data.

Apart from adaptive learning for ensemble models, other papers have studied the ways of adaptively selecting informative instances to re-sample from and learning the best sample rate automatically (Lu et al., 2016; Moreo et al., 2016; Ren et al., 2016a; Yun et al., 2016; Yeh et al., 2016; Zhang et al., 2016a). Furthermore, Krawczyk et al. (2014) tried to learn cost matrix of cost-sensitive learning from data. Noting that these are all most recent studies, which also support that adaptive learning could be another research topic for imbalanced learning.

#### 5.1.3. Online learning for imbalanced data stream classification

The sheer volume and accessibility of data draws much enthusiasm to big data analytics; one of its challenges is to tackle and respond to streaming and fast-moving input data. Online learning, which aims to process one example at a time, thus has gained increasing attention in data mining community. First, it receives an example and then makes a prediction. If the prediction is wrong, it suffers a loss and updates its parameters (Maurya et al., 2015). Skewed class distributions can be seen in many data stream applications, such as in the fault diagnosis of control monitoring systems and intrusion detection in network and spam identification (Hoens et al., 2012). When learning data streams online, three main difficulties may arise: a). the underlying data distribution often changes considerably over time, which is referred to as concept drift (or non-stationary) learning (Ghazikhani et al., 2014). b). Online class imbalanced learning has significant difficulties because there is a lack of prior knowledge about which data classes should be regarded as the minority or the majority and the uncertainty imbalance status (Wang et al., 2014b, 2015a). c). Data sparsity problem is commonly found in data streams (Maurya et al., 2016). These encourage research into dynamically determining the class imbalance status in data streams and effectively adapting online learners to the class imbalance (Ghazikhani et al., 2013a).

Boyu Wang (2016), Dal Pozzolo et al. (2015), Ditzler and Polikar (2013), Wang et al. (2013) designed ensemble models with re-sampling for learning imbalanced data streams. However, cost-sensitive was rarely seen in the existing literature, only three models are found in Ghazikhani et al. (2013b), Maurya et al. (2016) and Wang et al. (2014a). When classifying big data streams, cost sensitive learning is computationally more efficient than data sampling techniques. We thus recommend researchers pay more attention to cost-sensitive online learning. Moreover, ELM based online learning algorithms have gained popularity, as the efficiency of ELM meets the real-time prediction requirement (Mao et al., 2017; Mirza et al. 2015a,b)." Since the requirements of quick and accurate responses for any data that may arrive at any time is increasing in the era of big data, online learning in a dynamic and imbalanced scenario may become a popular new research topic.

#### 5.1.4. Semi-supervised learning and active learning

In some data analysis domains, massive data is cheap to collect; however, it is expensive to obtain labeled examples to train a classifier. Massive corpora with a few labeled instances (typically a minority) and abundant unlabeled instances are common in big data. Semi-supervised learning techniques have attempted to leverage the intrinsic information in unlabeled instances to improve classification models (Zhu and Goldberg, 2009); however, these techniques have assumed that the labeled instances cover all the learning classes, which often are not the case. Moreover, when there is imbalanced class distribution, extracting labeled instances from minority classes might be very costly. One way to gather more labeled examples is to ask experts or users for extensive labeling, which could lead to a specific semi-supervised learning method, called active learning. Active learning allows an expert to label new instances based on criteria to reduce the labeling effort (Frasca et al., 2013). The general idea in active learning is to estimate the value of labeling one unlabeled instance. The most valuable queries, given the goals of the classification task, are selected by the learning algorithm instead of being randomly selected as is the case in passive supervised learning (Escudeiro and Jorge, 2012). Few active learning algorithms have been proposed to address imbalanced data (Dong et al., 2016; Fu and Lee, 2013; Oh et al., 2011). More efforts are needed to investigate the selection

and utilization of informative examples when an imbalanced data distribution exists.

### 5.2. At a practical level

Revisiting the application distribution summarized in Section 4, two research domains closely related to management science and decision-making had adopted few imbalanced learning techniques. The first was emergency management. Four references were found that sought to predict natural disasters under imbalanced distributed data, as natural disasters are typical rare events. However, other types of emergency events, including accidents (such as forest fires), public health incidents (outbreaks of diseases such as cholera, Ebola, and malaria) and social security incidents (such as terrorist attacks) were rarely discussed in the imbalanced learning area. With the development of the Internet of Things, affluent monitoring data collected by sensors is accessible to researchers and scientists. Since large scaled multi-source and heterogeneous data can be easily collected in the big data era, it may be probable to develop feature engineering techniques to fuse multi-source data such as sensor data, text on the Internet and surveillance videos to build machine learning systems to detect other types of emergency events. Imbalanced learning techniques are crucial when designing learning models as the collected data related to emergency events may be imbalanced.

Another valuable research direction, from our point of view, is adapting imbalanced learning to security management issues. In particular, internet security management has had increased attention in recent years because of the rapid development of social networks. People tend to express their loves and hates on anything from a movie to a political strategy in social media, which also makes it possible for extremists to offend public order. Sentiment analysis and rumor detection may be powerful methods to monitor social networks and prevent the occurrence of risky events. Mining risk statements in massive user generated content is a rare event detection problem, which could be solved using imbalanced learning techniques.

## 6. Conclusions

In this paper, we attempted to provide a thorough review of rare event detection techniques and its applications. In particular, a data mining and a machine learning perspective was taken to view rare event detection as a class imbalanced data classification problem. We collected 527 papers that are related to imbalanced learning and rare event detection for this study. Unlike other surveys that have been published in the imbalanced learning field, we reviewed all papers from both a technical and a practical point of view.

Through our review, we also found some insights about commonly-used methods in some domains:

1) In chemical and biomedical engineering areas, re-sampling based ensemble classifiers are widely-employed. Since data that used in these areas are usually clinical data with fixed structure, feature engineering is rarely considered. However, for those high-dimensional data (such as protein data), feature selection is a popular choice.
2) A sophisticated feature engineering process is important for some management tasks such as financial management and business management. The features used for coping with a specific task are usually well-designed by the experts. Different from other domains, the goal of prediction in such field is often profit-driven instead of accuracy-driven. Therefore, cost-sensitive learning is often utilized and the cost of misclassification can be decided by experts or managers. Some widely-used classifiers in management fields are rule-based classifiers, such as decision tree and expert systems, in which fuzzy theory is often incorporated. This may due to that, apart from making wise decisions, understanding the criteria of decision-making is also essential for companies.
3) The main challenge of rare events detection in IT is the complexity of the data. Network log and unstructured data such as text and image usually need data cleaning and feature engineering processes. Besides, data streams are widely existed in IT area, which require online learning instead of traditional offline learning.

At the last of this paper, we incorporated some future suggestions from the reviewed papers with our thoughts to propose some future research directions for imbalanced learning and rare event detection, which will also be the focus of our future research projects.

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